Rahul Verma

+91 9525113211 rahulverma.work1@gmail.com github.com/RahulVerma32 linkedin.com/in/rahul-verma-connect

Education

Indian Institute of Information Technology, Bhopal Nov 2022 - June 2026

B.Tech. in Electronics and Communication Engineering *GPA: 9.00/10*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Experience

314e Corporation — SDE Intern January 2026 – July 2026

Software Development Intern *Onsite – Bengaluru*

- Designed and built an **asynchronous bulk-export pipeline** (ZIP/CSV) processing **100+ documents per day**, using concurrent S3 downloads to reduce export time by **30%** and a task queue to decouple long-running jobs from the request path.
- Built a **configurable metadata engine** using CEL expressions for dynamic export columns, backed by PostgreSQL JSONB aggregation for export history, and integrated Novu for real-time success and failure notifications.
- Built **full-stack bulk task assignment and cancellation** with batch SQL authorization and parallel execution, reducing bulk task processing time by **20%** and improving productivity for large-scale operations.
- Delivered an end-to-end **User Group management system** with full CRUD, a SQL JOIN-level active filter for task resolution, and trigger-backed audit history.
- Led the end-to-end **notification pipeline migration** by upgrading the Novu SDK within Temporal asynchronous workflows and shipping **organization-level and per-user Slack OAuth integration**.

Projects

AI-Powered Interactive Story Game GitHub | Demo

FastAPI, React, LLM APIs, LangChain, OpenRouter, PostgreSQL, Docker,

- Developed an **LLM-powered interactive storytelling engine** using structured prompting to generate **3-chapter branching narratives** with 10 scenes per chapter and 3 choices per decision.
- Built a **stateful REST API** with FastAPI, SQLAlchemy, and PostgreSQL to persist stories, player choices, and progress; implemented a 3-heart game mechanic and secured user access with OAuth 2.0 and JWT authentication.
- Created a React frontend with randomized choices and asynchronous API integration; containerized the backend with Docker and configured GitHub Actions CI/CD with **17 backend tests** for deployment on Render and Vercel.

ChatMitra GitHub | Demo

MongoDB, Express.js, React.js, Node.js, Socket.IO, WebRTC, Cloudinary, JWT, Bcrypt

- Built a **full-stack real-time chat platform** using the MERN stack and Socket.IO, supporting **100+ concurrent connections** and instant message delivery.
- Integrated **WebRTC video calling** and real-time features including typing indicators, read receipts, presence tracking, and push notifications; optimized Cloudinary uploads to reduce payload size by **60%**.
- Implemented **secure authentication and API protection** using JWT, bcrypt, httpOnly cookies, SameSite policies, input validation, and rate limiting.

Achievements & Honors

- Served as a **Teaching Assistant at IIIT Bhopal** (July 2023 – Dec 2023), guiding **150+ students** in **Fundamentals of Computer Science**.
- Solved **400+ Data Structures and Algorithms problems** across platforms (**LeetCode**, **GeeksForGeeks**).
- Achieved a **3-star rating** on **CodeChef** with a **maximum rating of 1603**.
- Achieved a maximum rating of **1558** on LeetCode, ranking in the **top 29.94%** globally.
- Secured Global Ranks of **197** and **210** out of **10,000+** participants in CodeChef Starters **197** and **212** (Div-3).

Skills

- **Languages:** C/C++, Python, JavaScript, TypeScript, SQL
- **Frameworks:** React.js, Node.js, Express.js, FastAPI, Tailwind CSS, Bootstrap
- **Libraries:** TanStack Query, Axios, React Router, Material-UI (MUI)
- **Databases:** PostgreSQL, MongoDB, MySQL